## Comparative Analysis of Selected Research Papers

| SL No. | References | AI/ML Technique | Healthcare Application | Dataset Used | Key Contributions | Publisher |
|---|---|---|---|---|---|---|
| **1** | Lo et al. (2021) [16] | CNN, Federated Learning | Diabetic Retinopathy Classification & Vessel Segmentation | OCT retinal images | First privacy-preserving FL framework for OCT segmentation and DR classification across institutions | Elsevier (Ophthalmology Science) |
| **2** | Polap et al. (2021) [17] | CNN + Federated Learning + Blockchain | Skin Cancer Detection | HAM10000 | Integrated blockchain with FL to improve privacy and secure medical image sharing | Elsevier (Journal of Information Security and Applications) |
| **3** | Feki et al. (2021) [18] | VGG16 + FL | COVID-19 Detection | Chest X-ray | Demonstrated collaborative COVID-19 diagnosis without sharing patient data | Elsevier (Applied Soft Computing) |
| **4** | Polap & Woźniak (2021) [19] | Meta-heuristic FL | Skin Lesion Classification | HAM10000 | Proposed meta-heuristic aggregation to improve federated model optimization | Elsevier (Applied Soft Computing) |
| **5** | Li et al. (2020) [20] | Deep Neural Network + FL + Domain Adaptation | Autism Spectrum Disorder | ABIDE fMRI | Multi-site privacy-preserving brain MRI analysis with differential privacy | Elsevier (Medical Image Analysis) |
| **6** | Yang et al. (2021) [21] | Semi-supervised FL | COVID-19 CT Segmentation | Multi-national Chest CT | Cross-country federated segmentation for COVID lesions | Elsevier (Medical Image Analysis) |
| **7** | Polap (2021) [22] | Fuzzy Consensus + FL | Skin Disease Classification | HAM10000 | Introduced fuzzy consensus mechanism for federated decision making | IEEE Access |
| **8** | Chakravarty et al. (2021) [23] | ResNet18 + FL | Chest Radiograph Screening | CheXpert | Site-aware federated learning for multi-center chest disease detection | IEEE (ISBI) |
| **9** | Zhang et al. (2021) [24] | Dynamic Fusion CNN + FL | COVID-19 Detection | Chest X-ray & CT | Dynamic feature fusion to improve federated COVID diagnosis | IEEE Internet of Things Journal |
| **10** | Qayyum et al. (2022) [25] | Multi-modal FL | COVID-19 Diagnosis | X-ray + Ultrasound | Edge-based collaborative diagnosis using multiple imaging modalities | IEEE Open Journal of the Computer Society |
| **11** | Linardos et al. (2022) [26] | ResNet18 + FL | Cardiomyopathy Diagnosis | Cardiac MRI | Simulated multi-center federated imaging diagnosis with strong AUC performance | Nature (Scientific Reports) |

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| **12** | Kaissis et al. (2021) [27] | Deep CNN + Secure Aggregation + FL | Pneumonia Detection | Chest X-ray | End-to-end privacy-preserving medical imaging using secure aggregation | Nature Machine Intelligence |
| **13** | Adnan et al. (2022) [28] | Multiple Instance Learning + FL + Differential Privacy | Lung Cancer Detection | TCGA Cancer Data | Combining FL with differential privacy for secure pathology analysis | Nature (Scientific Reports) |
| **14** | Yan et al. (2021) [29] | ResNet18 + FL | COVID-19 Detection | COVIDx | Experimental evaluation of FL on COVID imaging across distributed clients | Springer |
| **15** | Hashmani et al. (2021) [30] | Adaptive Federated Machine Learning | Skin Disease Detection | ISIC 2019 | Intelligent dermoscopy system using adaptive FL | MDPI (Applied Sciences) |
| **16** | Zhou et al. (2022) [31] | Communication-efficient FL | Diabetic Retinopathy Classification | DR dataset Retinal images | Reduced communication overhead while maintaining classification accuracy | SPIE |
| **17** | Salam et al. (2021) [32] | Federated Machine Learning | COVID-19 Detection | Chest X-ray | Demonstrated distributed COVID diagnosis with competitive performance | PLOS ONE |
| **18** | Pfitzner et al. (2021) [5] | Systematic Review | Medical Federated Learning | Multiple Studies | Comprehensive review of FL in healthcare with privacy emphasis | ACM Transactions on Internet Technology |
| **19** | Prayitno et al. (2021) [6] | Systematic Review | Healthcare Federated Learning | Multiple Studies | Reviewed FL applications, datasets, privacy mechanisms, and challenges | MDPI (Applied Sciences) |
| **20** | Rieke et al. (2020) [12] | Federated Learning Framework | Digital Health & Medical Imaging | Multiple Clinical Datasets | Landmark review discussing FL opportunities, challenges, and future healthcare applications | Nature Digital Medicine |

**Summary of Comparative Analysis:**

This table includes 20 representative references selected from the 95 references cited in the reviewed survey paper. The selection was limited to publications from IEEE, ACM, Elsevier, Springer, Nature, and MDPI to ensure high-quality sources. These studies collectively represent the major AI/ML techniques, healthcare applications, datasets, and significant contributions in Federated Learning-based medical image analysis.